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STUDY OF HYDROGEN ISOTOPES SUPERPERMEATION THROUGH VANADIUM MEMBRANE ON "PROMETHEUS" SETUP

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To develop the membrane pumping technology by means of superpermeable membranes at RFNC-VNIIEF in the "Prometheus" setup, the experiments on superpermeation of hydrogen isotopes through metal membranes were carried out.

The experimental results on superpermeation of thermal atoms of hydrogen isotopes including tritium through a cylindrical vanadium membrane are presented. The possibility of effective pumping, compression and recuperation of hydrogen isotopes by means of superpermeable membrane was demonstrated. The evaluation of membrane pumping rates and asymmetry degree of pure vanadium membrane was given.

The work was performed under the ISTC-2854 project.

I. INTRODUCTION

Selective pumping and recycling techniques based on superpermeability can be employed in the fusion reactor fuel cycle and periphery systems [1,2]. The authors suggest that a major part of the D/T (e.g. 99%, or even more) should be extracted from the reactor exhaust and put back into the fuel cycle before it reaches the cryopumps, which are designed for evacuating virtually only helium and gas impurities. The fuel mixture is to be isolated by means of superpermeable membranes placed on the path between the core plasma and the cryopumps; such membranes are capable of selectively letting through hydrogen (D/T) – continuously, with a very high speed, and with an automatic orders-of-magnitude compression for the permeating gas.

The materials most suitable to obtain superpermeability is the *Group 5 metals*: V, Nb and Ta. *Vanadium* is likely to be the best of them for fusion applications due to its uniquely low induced radioactivity. However, detailed investigations undertaken so far mostly pertain to niobium [3], data on the superpermeability of vanadium being very scant.

II. EXPERIMENTAL

Experimental studies of the superpermeability phenomenon have been performed with a cell (Fig.1) of the "Prometheus" setup [3, 4]. Similarly to experiments with niobium [3] the cell comprises a cylindrical vanadium membrane (100 mm diameter, 180 mm height, and 0.1 mm thickness) and a thermal atomizer. The membrane is positioned vertically to divide the cell into an inlet and an outlet chamber. The inlet and outlet chambers are continuously pumped with ion pumps through ducts with conductivities of 18 l/s and 15 l/s for H_2, respectively. The thermal atomizer placed inside the membrane consists of three separate tantalum ribbons that can be ohmically heated up to a temperature of 2300 K. Their total surface area amounts to 120 cm^2 (about 20%

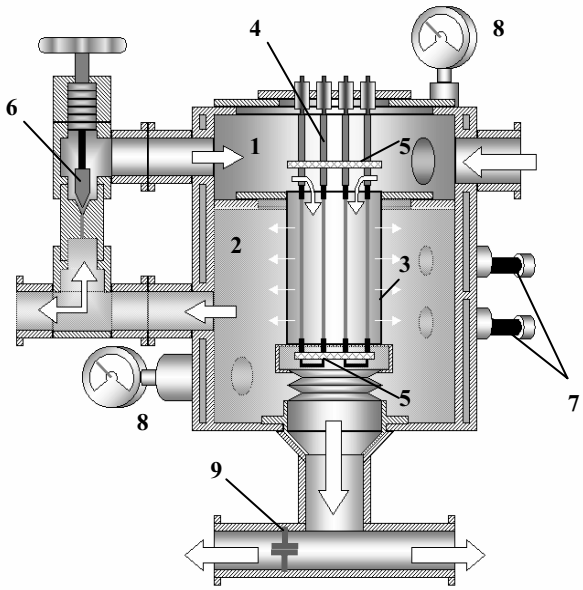


Fig.1. The cell of "Prometheus" setup for study of superpermeation of hydrogen isotopes: 1- inlet chamber; 2- outlet chamber; 3- cylindrical membrane; 4- atomizer; 5- quartz disks; 6- leak valve; 7- pyrometer; 8- pressure gauge; 9- diaphragm

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of the membrane area).

The membrane temperature is measured with pyrometers. The cylindrical membrane ends are screened by two quartz disks to prevent the losses of atoms produced by the atomizer inside the membrane hollow, and also to partly protect other device elements from the influence of atomic hydrogen.

III. MEMBRANE PUMPING AND DEMONSTRATION OF COMPRESSION AND TRITIUM RECIRCULATION

First a constant flux of hydrogen is established through the continuously pumped inlet chamber at pressures between 10^{-7} and 10^{-4} mbar. In order to prevent embrittlement of the atomizer ribbons because of hydrogen absorption, their temperature was kept in the range of 500K to 700K. Heating of the atomizer ribbons to temperatures between 1300K and 2100K results in a drastic rise in the dissociation rate (atomization) of hydrogen molecules at the atomizer surface. These atoms strike the membrane inlet surface, stick to it, dissolve in the metal, permeate through the membrane, and are released from the downstream surface as molecules. Most of the atoms that are produced permeated through the membrane in our experimental conditions. As a result we obtain a typical signature of the superpermeability phenomenon: the pressure in the inlet chamber is decreased steeply, while the pressure of hydrogen in the outlet chamber rises until a steady state permeation flux is established (Fig. 2). Reducing the atomizer temperature to its initial value ("switching off" of the atomizer) causes the inlet and outlet pressures to return to their initial levels.

A special experiment has been carried out to demonstrate a possibility of compression and recirculation of permeating hydrogen. The pumping of the outlet chamber with the external pump was ceased after permeating flux reached the steady state-level. As a result, the outlet pressure increased (see Fig. 3) due to the accumulation of permeating hydrogen in the outlet chamber (compression mode). That was accompanied with an inlet pressure rise because of enhanced molecular driven permeation through the membrane back from the outlet to the inlet side. Comparison of the pressure increment in the input and output sides was used to calculate the important membrane parameter of maximum compression (ratio of outlet to inlet pressures). As one can see in Fig.3, the compression value exceeds three orders of magnitude.

The obtained data also demonstrated that the membrane behaves strongly asymmetric – a major part of atomic hydrogen absorbed by membrane at the inlet side is released to outlet side. Comparison of the pressure increment in the input and output sides permitted the calculation of the degree of the membrane asymmetry

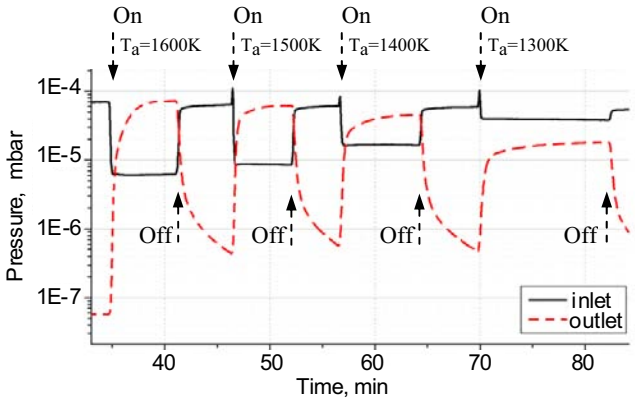


Fig.2. A typical pattern of hydrogen isotopes pressures at inlet and outlet sides of the membrane by "switching on" the atomizer to different temperature.

(relation between the inlet permeating flow and the outlet flow), which in turn depends on the sticking coefficients (experimental value: $\alpha_{inlet}=2\cdot10^{-5}$ to $5\cdot10^{-5}$ and α_{outlet} $\sim2\cdot10^{-4}$). The probable cause of the asymmetry is deposition of tantalum onto the inlet membrane surface from atomizer.

A membrane pumping system based on superpermeability permits an effective method of separate helium and fuel mixture from the machine exhaust. Since the membrane is able to compress the gas permeating through it, one also can arrange a direct circulation of the hydrogen evacuated by the membrane to return into the fuel cycle.

In order to demonstrate the possibility of circulation a leak valve (6 in Fig. 1) connecting the outlet and inlet chambers was opened to let part of the tritium gas

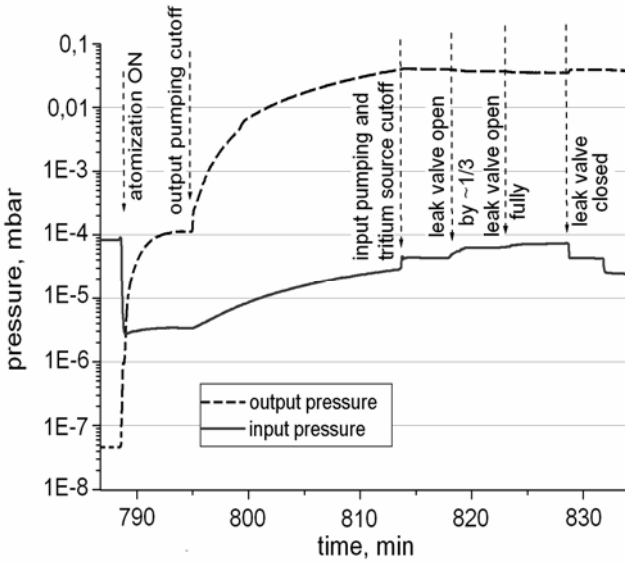


Fig.3. Demonstration of compression and tritium recirculation by means of superpermeable vanadium membrane.

accumulated in the outlet chamber back into the inlet chamber. Before the valve was opened (Fig.3) both the inlet chamber ion pump and the tritium admittance from tritium source were stopped. Varying the vacuum conductivity of the leak valve changed the influx so that the inlet pressure could be adjusted to any value in the operation range. As a result, a closed tritium circuit was formed.

The membrane pumping rate for hydrogen isotopes was determined at different temperatures of atomizer by comparing it with the known duct conductivities of the inlet chamber evacuated by the external ion pump. As shown in our previous paper devoted to niobium membrane performance [4], the selective pumping rate of the superpermeable membrane is inversely proportional to the square root of the isotopic mass as a first approximation. Experimental data on the membrane

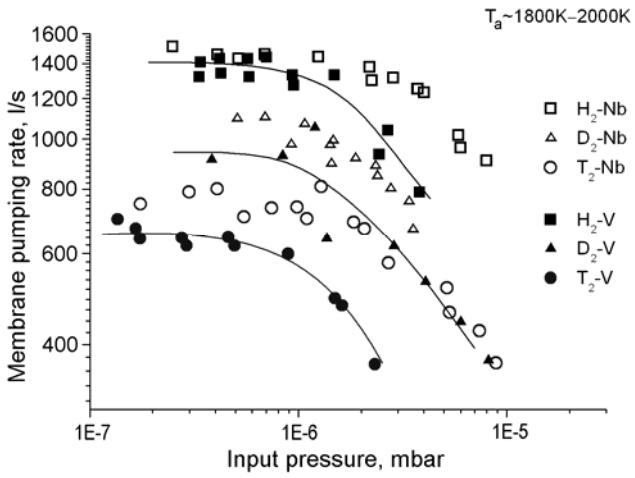


Fig.4. Comparison of the membrane pumping rates for protium, deuterium and tritium

pumping rate of protium, deuterium and tritium as a function of inlet gas pressure are shown in Fig.4. The maximum pumping rate amounts to ~2.3, 1.6 and 1.2 l/(cm^{2}sec) of the membrane surface for protium, deuterium and tritium respectively. Data for niobium membrane are also given here for comparison [3].

IV. CONCLUSION

The superpermeation of hydrogen isotopes, including tritium, through vanadium membrane was experimentally measured. The membrane pumping rate was found to be inversely proportional to square root of isotopic mass.

The possibility of effective pumping, compression and circulation of tritium by means of superpermeable vanadium membranes as well as niobium membranes was demonstrated.

ACKNOWLEDGMENT

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REFERENCES

[1] A.I. LIVSHITS, M.E. NOTKIN AND A.A. SAMARTSEV, "Physico-chemical origin of superpermeability – large-scale effects of surface chemistry on "hot" hydrogen permeation and absorption in metals," *J.Nucl.Mater*. 170, 79–94 (1990).

[2] A. I. LIVSHITS et al., "Superpermeability: Critical points for Applications in Fusion", *J.Nucl.Mater.* **220-222**, 259 (1995).

[3] R.K.MUSYAEV et al., "Tritium superpermeability: experimental investigation and simulation of tritium recirculation in "Prometheus" setup", *Fusion Science and Technology*, **48**, 35-38 (2005).

[4] A.A.YUKHIMCHUK et al., "Device for investigations of tritium retention in and permeation through metals and structural materials,". *Hydrogen Recycling at Plasma Facing Materials, NATO Science Series, II Mathematics, Physics and Chemistry*, **1**, 339-348 (2000).

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